

Five Business Days: Activist Disclosure Deadlines, Stake Formation, and the Market for Corporate Control

A research outline

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Abstract

In February 2024 the SEC shortened the Schedule 13D filing window from ten calendar days to five business days, while dropping a proposed rule covering cash-settled derivatives. We will provide the first realised evidence on this reform. We will construct a predetermined, target-level exposure measure — the trading days required to move five per cent of shares outstanding at a liquidity-capped pace, which fixes how much undisclosed accumulation the window permits once the five-per-cent threshold is crossed and the statutory clock starts — and test whether campaign entry and composition respond to the new deadline in proportion to exposure, and whether activists substitute equity accumulation with derivatives disclosed under Item 6, adjudicating the SEC’s \$810 million annual foregone-value projection. A deadline-accumulation model with a liquidity-based price-impact microfoundation generates the exposure measure and organises the predictions; the deadline binds as a corner solution for a model-generated subset of targets. We will further decompose activist target gains into filing returns, post-filing run-ups, and bid markups, stated event-time-invariant, to establish where takeover value reaches target shareholders. All designs carry pre-registered minimum detectable effects and use the SEC’s published tables as the pre-period anchor.

1 Research Question

In February 2024 the Securities and Exchange Commission accelerated the initial Schedule 13D filing deadline from ten calendar days to five business days, the first change to the accumulation window since 1968 (Securities and Commission 2023). The amendment compresses the time an activist has to buy shares quietly between crossing the five-per-cent threshold and disclosing the position, at the same time as the final rule declined to adopt proposed Rule 13d-3(e), which would have treated certain cash-settled derivatives as beneficial ownership. The main question of this paper is: did accelerating mandatory ownership disclosure change who launches activist campaigns, how activist stakes are formed — including any migration of exposure into cash-settled derivatives that the final rule chose not to cover — and how takeover value reaches target shareholders? We organise this into three sub-questions.

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1. **Entry and composition.** Did the acceleration reduce the number of activist campaigns (the extensive margin), and did it shift the composition of campaigns towards more liquid targets, towards larger activists, or towards different campaign intents? The identifying variation is predetermined target-level exposure: the number of trading days required to move five percentage points of ownership at a liquidity-capped pace, $RTD_i = (0.05 \times S_{out,i})/(\phi \times ADV_i)$, with $S_{out,i}$ shares outstanding. The statutory clock starts only when the activist crosses the five-per-cent threshold; over the following d business days the window permits an undisclosed increment of at least another five percentage points if and only if $RTD_i \leq d$. RTD_i is measured from a lagged pre-trigger window and does not depend on the campaign’s realised accumulation or outcome.
2. **Stake formation and substitution.** Among campaigns that still occur, did stakes shrink, did accumulation shift earlier or later around the trigger date, and did activists substitute equity accumulation with cash-settled derivatives disclosed under Item 6 — a margin the final rule left uncovered when proposed Rule 13d-3(e) was dropped? Substitution predicts that exposure concentrates in precisely the high- RTD_i targets where the deadline binds.
3. **Gain timing and anatomy.** Where do target shareholder gains actually land? Decomposing the total return into the filing return $G_{fil} = \log(P^F/P^0)$, the post-filing run-up $G_{run} = \log(P^B/P^F)$, and the bid markup $G_{mkp} = \log(B/P^B)$, we ask whether disclosure policy shifts value between these components even when it leaves their sum unchanged, and whether activist-involved targets receive lower markups because the market has already capitalised the campaign.

2 The Conventional Wisdom This Paper Challenges

2.1 Myth 1: faster disclosure must chill activism

The policy debate has already decided the answer. The SEC’s own economic analysis projected that the acceleration would impose costs on activists of roughly \$810 million per year in foregone shareholder value, on the argument that a shorter window raises the cost of the undisclosed position an activist can add once the five-per-cent threshold is crossed and therefore deters value-increasing campaigns (Securities and Commission 2023). Polk et al. (2024) project reduced activist returns from historical data, and practitioner commentary at adoption predicted the effective death of the accumulation window. Yet the same regulatory tables that motivate the projection show that roughly 80 per cent of filers already completed their accumulation within the new five-business-day deadline, and the window binds materially on only a small minority of campaigns. If most activists never used the old window, the projected chilling may never materialise; and if constrained activists responded not by exiting but by moving exposure into cash-settled swaps through the gap left by the dropped Rule 13d-3(e), the reform may have changed the *form* of activist ownership rather than its amount. Whether the reform chilled anything, and whether the response migrated into derivatives, is an open empirical question that, to date, no study has answered with realised post-implementation data. Existing evidence is either pre-reform projection (Polk et al. 2024) or drawn from foreign disclosure regimes (Bonetti et al. 2020), which carry their own institutional baggage.

2.2 Myth 2: activists sell targets cheap

A second piece of conventional wisdom holds that activist involvement lowers takeover premia, and hence that activists sell their targets too cheaply. Celentano and Levine (2025) document that conditional takeover premia are indeed lower for activist-targeted firms, and the natural reading — that the activist’s presence weakens the target’s bargaining position — has entered the policy conversation. But the measured premium is a ratio whose denominator is the pre-bid price, and that price already capitalises the campaign: the market learns that a sale is more likely from the 13D filing itself and from post-filing trading, so part of the takeover gain is paid to shareholders before any bid is announced. Whether run-ups substitute for or add to the final offer is itself contested: Schwert (1996) found little substitution — the run-up was largely *added* to the bid — while Betton et al. (2014) show that once run-ups are adjusted for the target’s stand-alone value change, run-up and markup do move against each other. The adjusted substitution result implies that measured markups can fall while total target gains rise. We treat this capitalisation argument as a measurement lemma rather than a headline claim, and we test it directly: the gain anatomy shows where gains land, in filing returns, run-ups, and markups, for activist-involved and matched targets, stated event-time-invariant so that the reform itself cannot mechanically truncate the window and mimic the result.

2.3 Contribution

Existing theory models disclosure thresholds as static size triggers (Ordonez-Calafi and Bernhardt 2022), the activist as a broker or intermediary with an exogenous toehold (Burkart and Lee 2022), control contests with instantaneous solicitation and disclosure (Corum and Levit 2019), and structural quantification of campaign and sale outcomes (Johnson and Swem 2021; Albuquerque et al. 2022; Celentano and Levine 2025), but no paper models the filing *window* as a time margin of stake formation or measures the realised response to the 2024 acceleration. This paper’s contribution is the realised evaluation: a chilling/substitution dose-response design on predetermined required trading days. The deadline-accumulation model serves only as the lens that generates the exposure measure and organises the predictions.

3 Literature and Position

This section positions the paper in four literatures — disclosure-threshold theory, the theory of activism and the market for corporate control, the structural quantification of activism, and the measurement of takeover gains — and then states the reform and the novelty claim. The characterizations below are disciplined by a machine-checked reading of the full texts; where a claim rests on a verbatim quotation, the quotation is logged in the accompanying verification report. The through-line is that the activist-as-intermediary position is occupied at the theory level, while two margins are open: the filing *window* as a time margin of stake formation, and *realized* evidence on the February 2024 acceleration.

3.1 Disclosure-threshold theory

Grossman and Hart (1980a) is the classical antecedent: mandatory disclosure of a building position raises the acquirer’s cost of stealth and so trades off the protection of target shareholders against the disciplining of management. The modern theoretical treatment is Ordonez-Calafi

and Bernhardt (2022), who study a blockholder disclosure threshold as a *static stake-size cap*: in a one-shot Kyle-style dealership trade, a competitive market maker posts prices against net order flow, and the activist chooses a single position once. Their policy instrument is an ownership level — a shareholder must disclose once holdings reach a given fraction of voting rights — not a time margin. There is no accumulation period, no filing deadline, and no second agent representing a potential bidder; the welfare result is the familiar transparency-versus-discipline trade-off, in which uninformed investors favor lower thresholds when their trading losses against the informed activist are small relative to the activist’s disciplining benefit. Critically for us, their Section V explicitly gestures at objective-specific thresholds — that the right disclosure rule may depend on what the activist is trying to do, including sale-seeking campaigns — but leaves this as an open question. Answering it, with the filing window rather than the stake level as the lever, is part of what this paper does.

Bonetti et al. (2020) is the structurally closest *design* paper, and it is empirical rather than theoretical. Using the staggered country-level implementation of the E.U. Transparency Directive (2007–2009) as a difference-in-differences shock, they show that tighter ownership disclosure *shrinks toeholds, reduces the number of control acquisitions, and moves announcement returns in opposite directions for target and acquirer*. Their setting is ordinary corporate bidders building pre-bid toeholds, not hedge-fund activist 13D campaigns, and their reform bundles a lower ownership threshold with a shorter notification deadline, so it is not an isolated window shock the way the 2024 Schedule 13D change is. The reason to engage them head-on is the *sign* of the bidder-side channel. Building on Fishman (1988), their bidding-competition mechanism has disclosure *detering* rival bidders: once alerted, competitors can make competing bids that raise the acquirer’s cost. That is the opposite sign from the certification channel this paper emphasizes, in which earlier disclosure *attracts* bidders by signaling that an activist-flagged target is available for sale. The two signs are not contradictory; they are the two sides of a single elasticity. Our certification-versus-deterrence result (Proposition R4) nests both: the bid hazard rises after earlier disclosure if and only if the certification elasticity exceeds the deterrence elasticity, and the observable heterogeneity that signs it (sale-intent filings, opaque assets with few natural buyers, high industry merger intensity) is precisely what separates an activist-flagged sale from the ordinary toehold acquisitions Bonetti et al. (2020) study.

3.2 Activism and the market for corporate control

The theoretical position this paper builds on — the activist as an intermediary between an underperforming target and the takeover market — is already occupied, and we say so plainly. Burkart and Lee (2022) make it their headline: in their words, “activism is most efficient when it brokers, rather than substitutes for, takeovers.” Their activist runs a proxy-style campaign for control of the board and then uses the board’s merger authority to sell the whole firm to a third-party bidder, mitigating both the Grossman–Hart and Jensen–Meckling free-rider problems. But the toehold is exogenous and the model is entirely post-disclosure: there is no stake accumulation and no window. Their concluding remarks explicitly defer “the acquisition of the toehold in anonymous, predisclosure markets” to future work — the margin we open.

Corum and Levit (2019) occupy the bidder-discovery side of the same position. In their model bidders “interpret the presence of an activist as a signal that the target is available for sale,” so the activist’s presence puts the target in play and raises the probability of a takeover. It is important to be precise about what they do and do not model, because an earlier draft overstated this. They *do* model secondary-market inference: the activist builds a position by

trading against a Kyle (1985) market maker who learns from order flow, and the activist’s ownership then becomes public through a Schedule 13D publication step. What they do *not* model is a disclosure *regime* — the publication step is instantaneous, with no threshold design and no filing window or deadline to vary. Our contribution relative to them is therefore narrow and honest: the window-as-deadline time margin, and the realized reform, not the claim that they ignore trading. Their footnote 29 also already delivers the sign often credited to later structural work: when the activist’s proxy threat is credible, a takeover is more likely but the bidder pays a lower conditional premium, because the activist weakens the board’s negotiating position.

Cetemen et al. (2026) is a continuous-time leader–follower model of how a leader activist’s trades tacitly coordinate followers, producing wolf-pack accumulation without formal agreement. It is pure theory about coordination through trading, and it is *not* silent on the reform — a footnote cites the February 2024 acceleration (ten calendar days to five business days) as institutional background — but it does not incorporate the window as a variable or analyze it. The honest claim is therefore that no theory paper gives the window *analytical* treatment, not that the reform goes unmentioned.

Two strategic-trading papers are the benchmarks our reduced-form accumulation cost shortcuts. Back et al. (2018) endogenize the activist’s optimal trading and liquidity, and Collin-Dufresne and Fos (2016) solve the problem of an informed trader with long-lived private information revealed at a known date, trading against stochastic liquidity. A static quadratic cost $C(x; d, L) = (\lambda_L(L)/2) x^2/d$ is a fair reduced-form summary of the *level* effects these models deliver — more liquidity and a longer horizon lower the marginal cost of a given accumulation — but it collapses away their two dynamic contributions: endogenous intra-window timing (the insider trades more aggressively when noise trading is high) and stochastic, path-dependent liquidity. We use the reduced form to generate the exposure measure and organizing predictions, and we are explicit that a full dynamic treatment would add a timing margin the static cost cannot represent. The empirical reality that motivates a reduced form is Collin-Dufresne and Fos (2015): on an average day when Schedule 13D filers trade, measured price impact is almost 30% lower than the sample average, so standard liquidity proxies do not cleanly detect informed window trading — a caution for any design that uses pre-filing liquidity as a control.

The older blockholder literature supplies the background on exit, voice, and liquidity. Maug (1998) shows that liquidity raises the value of the exit option and so trades off against monitoring; Kahn and Winton (1998) argue that liquid markets can substitute speculation for intervention; Edmans et al. (2013) document that liquidity shifts blockholders away from voice and toward governance through exit; and Edmans and Holderness (2017) survey the theory and evidence on blockholders. None speaks to a disclosure window or to takeover-gain timing.

3.3 Quantifying activism: structural evidence

The structural literature quantifies campaigns but does not exploit disclosure timing. Johnson and Swem (2021) estimate a dynamic reputation model in which an activist’s track record for proxy fighting shapes targets’ responses across a sequence of campaigns. Accumulation is collapsed into a single reduced-form lognormal campaign-cost draw — round-trip liquidity costs are named as a verbal component of that one scalar but never separately estimated — and the filing window appears exactly once, in an institutional footnote describing the SEC’s ten-day rule. There is no mapping of the window to a noise or variance parameter and no counterfactual that varies filing delay; the model in fact has no separable disclosure-timing choice, because

filing the 13D is definitionally simultaneous with campaign initiation.

Albuquerque et al. (2022) study the choice between Schedule 13D and Schedule 13G and decompose the average 13D announcement return of 6.34% into treatment (75.2%), stock-picking (12.2%), and sample-selection (12.6%) components — a capitalization-at-disclosure result at the filing leg. Their activism cost is a single scalar held constant across filings, and they offer no takeover-premium counterfactual; an earlier project document misattributed the premium result below to them, and we correct that attribution here.

Celentano and Levine (2025) estimate a simulated-method-of-moments model on 2001–2019 data and report that, in takeovers with an activist present, “the bid premium is 13.7 percent (5.2 percentage points) lower than if the activist had not been present,” with the mechanism that the activist “lowers the negotiating position and price demanded by the board in negotiations” — reduced board bargaining leverage, not a liquidity or disclosure channel. Crucially, they themselves assert the capitalization reading of their own number: “the pre-announcement share price used to calculate the bid premium incorporates the activist’s signal about a potential acquirer.” Because the premium’s denominator already capitalizes the campaign, a lower measured premium need not mean less value to target shareholders. This is exactly why we treat capitalization as a measurement lemma rather than a headline: Celentano and Levine (2025) own the interpretation, and their model treats the activist’s decision and information as instantaneously common knowledge, with no disclosure timing anywhere, leaving the window and the realized reform open.

Gantchev (2013) estimates that the average activist campaign costs about \$10.5 million, roughly half of it in the proxy stage. These are the costs of contested campaign *tactics* — proxy-fight, legal, and advisory expenditure — and are conceptually distinct from the accumulation cost this paper’s model represents, the market-impact cost of quietly building a pre-disclosure stake. We cite the figure for scale, not as a calibration of our cost object.

3.4 Takeovers, run-ups, and the measurement of gains

The gain anatomy borrows the standard run-up/markup decomposition. Schwert (1996) writes the premium as the sum of a pre-bid run-up, measured as the target’s cumulative abnormal return from day -42 to day -1 , and a post-announcement markup. His headline finding, however, runs *against* substitution: he reports “little substitution between the runup and the markup,” with at least two-thirds of any run-up *added* to the total price the bidder pays. The substitution evidence our measurement lemma actually needs comes from Betton et al. (2014): once the run-up is adjusted for the target’s own stand-alone value change, isolating the deal-synergy component, the slope moves from -0.09 (following Schwert’s unadjusted method) to -0.39 , indicating “a substantially greater degree of substitution” and rejecting the mechanical costly-feedback loop. The run-up/markup window convention is standardized in the survey of Eckbo et al. (2025), which uses a run-up window of day -41 to day -2 and a separate three-day announcement window. Our three-part split — filing, run-up, and markup — adds an activist-filing-triggered leg that has no counterpart in the two-part bid-anchored convention, and we justify it on its own terms.

The activism–takeover link is a stylized fact. Greenwood and Schor (2009) show that the large abnormal returns to 13D activism are “largely explained by the ability of activists to force target firms into a takeover”: announcement and long-run returns are high only for targets ultimately acquired and indistinguishable from zero otherwise, and activism raises takeover probability by about eleven percentage points. Brav et al. (2008) report that selling the company is a stated

objective in roughly 14–18% of campaigns and that about 15% of activist targets are ultimately acquired. Boyson et al. (2017) study activism mergers directly and find that activist targets are far more likely to receive bids and to merge; our characterization rests on an author-posted summary rather than the primary text, so we keep it to this one conservative sentence and do not build our sale-intent codebook on it. None of these papers touches disclosure timing.

3.5 The reform and the novelty claim

The institutional shock is the SEC’s modernization of beneficial-ownership reporting (Securities and Commission 2023). The final rule shortens the initial Schedule 13D filing deadline from ten calendar days to five business days, effective February 5, 2024; requires amendments within two business days of a material change; and mandates XML-structured filing from December 18, 2024. Two features matter for the design. First, the rule *dropped* proposed Rule 13d-3(e): the Commission declined to deem holders of cash-settled derivative securities beneficial owners of the reference class, issuing guidance and an Item 6 disclosure clarification instead. That decision leaves a substitution hole — activists can migrate economic exposure into cash-settled derivatives without triggering a 13D — which is the second half of our chilling/substitution headline. Second, the SEC’s own baseline shows the reform binds a minority of campaigns: roughly 29% of all 2022 initial filings (about 41% of those filed on time) were already made within five business days, about 80% of filers had completed accumulation by the new deadline, and 97% of campaigns had reached 90% of their reported stake by it. The staff also projected foregone shareholder value of about \$810 million per year under the five-day rule — a projection this paper adjudicates with realized data. The February 2025 staff guidance (Securities and Exchange Commission 2025) clarifies when such engagement makes the shorter Schedule 13G unavailable (calls for a sale or restructuring, alternative director nominees, or conditional voting support), which bears on how sale-intent campaigns are routed between the two schedules.

The only prior study of the acceleration itself is Polk et al. (2024), a pre-reform projection on 9,685 initial 13D filings from 2001–2022 that compares investors who voluntarily filed within five days against those who waited the full ten. Its outcomes are abnormal returns and trading volume around the trigger and filing dates only: it contains no takeover, premium, bidder-entry, or campaign-resolution variables, and its sample ends more than a year before the reform took effect. A systematic novelty scan run on 2026-08-02 — spanning SSRN, NBER, arXiv, the SEC/DERA record, forthcoming top-journal pages, and author research pages — found *no* study of realized post-February-2024 Schedule 13D outcomes. The closest adjacent work, Celentano and Levine (2025), is not built around the reform and uses no post-reform data. The realized evaluation is genuinely open.

3.6 Position

The activist-as-intermediary position is taken at the theory level, and the capitalization reading of a lower conditional premium is asserted by its authors; we do not reclaim either as new. What remains open are two margins. The first is the filing window as a *time margin* of stake formation: disclosure-threshold theory fixes a static stake-size cap, the brokerage and solicitation models take the toehold as given or publish it instantaneously, and the structural literature collapses accumulation into a scalar cost draw, so no paper makes the legal deadline d a margin that binds only for a model-generated subset of targets. The second is *realized* evidence on the February 2024 acceleration: the existing study is a pre-reform projection with returns and volume

Table 1: Positioning. Columns: (Window) the filing window modeled as a time margin of stake formation; (Realized) realized evidence on the February 2024 Schedule 13D acceleration; (Anatomy) a decomposition of target gains into filing/run-up/markup; (13D) an activist Schedule 13D setting.

	Window	Realized	Anatomy	13D
Ordonez-Calafi and Bernhardt (2022)	no	no	no	yes
Burkart and Lee (2022)	no	no	no	yes
Corum and Levit (2019)	no	no	no	yes
Johnson and Swem (2021)	no	no	no	yes
Albuquerque et al. (2022)	no	no	partial	yes
Celentano and Levine (2025)	no	no	partial	yes
Bonetti et al. (2020)	partial	no	no	no
Polk et al. (2024)	partial	no	no	yes
Cetemen et al. (2026)	no	no	no	yes
<i>This paper</i>	yes	yes	yes	yes

only. This paper supplies both — a deadline-accumulation model used as a lens to generate the predetermined required-trading-days exposure $RTD_i = (0.05 \times S_{\text{out},i})/(\phi \times ADV_i)$ and to organize predictions, and a dose-response design on that exposure as the evidence. The event-time-invariant anatomy of target gains (filing, run-up, markup) is the measurement contribution that disciplines the capitalization lemma; the model is the lens, not the paper’s claim. Table 1 records where each paper sits.

4 Model: A Deadline Theory of Activist Stake Formation

This section builds the lens through which the empirical evaluation is read. It is deliberately small. Its job is to (i) *derive* the exposure variable RTD_i that the empirical design uses, rather than assert it; (ii) deliver a signed prediction for how a shortening of the filing window moves the undisclosed stake an activist can accumulate before filing; (iii) state the entry response as a testable inequality rather than a sign; and (iv) supply the exact measurement content of the gain decomposition, so that the second-act anatomy has a restriction to test and not merely an accounting identity to report. The capitalization interpretation of a compressed measured premium is *not* claimed here as a contribution: Celentano and Levine (2025) state and quantify that mechanism, and Betton et al. (2014) own the run-up/markup substitution evidence. What is new is the time margin. Neither Celentano and Levine (2025) (no disclosure rule) nor Ordonez-Calafi and Bernhardt (2022) (a *size* threshold) models the window as a deadline in an accumulation problem. Target: 10–14 pages in the paper; results are numbered R1–R6.

4.1 Environment and timeline

A single risk-neutral activist faces a single target firm i with S_{out} shares outstanding and liquidity L_i , measured empirically by average daily volume ADV_i over the six months ending 60 trading days before the trigger — lagged so the untimed toehold phase does not mechanically inflate the measure, aligned with the P^0 anchor — (or, equivalently in the cross-section, by Amihud illiquidity). All agents discount at $\delta \in (0, 1)$ per period. Dates:

- t_0 . **Opportunity arrival.** The activist privately draws an intervention opportunity $\theta \geq 0$: the per-share value improvement available if the campaign succeeds. The market price at a fixed horizon of k trading days before the trigger date t_1 is P^0 , the reference price. Dating P^0 at trigger-minus- k , not at filing-minus- k , keeps the anchor outside the statutory window, whose accumulation moves the price; because the pre-trigger toehold phase is untimed, purchases inside $(t_1 - k, t_1)$ can still contaminate P^0 — mechanically so where $\text{RTD}_i > k$ — so the empirical section diagnoses pre-trigger own-trading in the anchor window from the Item 5(c) sixty-day transaction tables and reports the 120-trading-day anchor as robustness.
- t_1 . **Trigger and accumulation window.** The activist first accumulates a toehold up to the statutory five-per-cent threshold at its own pace — the pre-trigger phase carries no deadline. Crossing the threshold is the *trigger date* (the adopting release’s own anchor) and starts the statutory clock: over the following d business days the activist buys the undisclosed increment x (a fraction of shares outstanding beyond the threshold stake). Pre-reform $d \approx 7$ business days (ten calendar days); post-reform $d = 5$.
- t_2 . **Mandatory filing** at $\min\{\text{deadline}, \text{voluntary filing}\}$. Early voluntary filing is permitted and is sometimes optimal; the legal deadline is therefore a constraint that binds on a model-generated subset, not on everyone. The post-filing price is P^F and $G_{\text{fil}} \equiv \log(P^F/P^0)$.
- t_3 . **Bidder entry.** Potential acquirers search and evaluate; the bid-eve price is P^B and $G_{\text{run}} \equiv \log(P^B/P^F)$.
- t_4 . **Bargaining and resolution.** A final offer B is agreed or the firm remains freestanding; $G_{\text{mkp}} \equiv \log(B/P^B)$.

One type dimension. The activist’s type is the scalar θ . We do *not* carry a second, operational-versus-sale type dimension. This is a deliberate restriction, and the reason is stated plainly: a two-dimensional type with a two-dimensional signal (x , Item 4 purpose) generically separates, P^F collapses to the full-information price, and the noise the rest of the model needs disappears. The operational-versus-sale distinction enters this paper only through the *empirical* Item 4 classification, where it indexes heterogeneity in the entry elasticities of R4 — it is not a second signalling dimension. See §4.8.

4.2 Accumulation technology

The activist trades against a competitive dealer sector with linear price impact $\lambda_L(L)$, decreasing in liquidity L (empirically, $\lambda_L \propto 1/\text{ADV}$ or $\propto \text{Amihud}$). Detection risk and the dealer’s inventory limit impose a participation cap: at most a fraction ϕ of ADV may be bought per day. Buying at daily rate r against linear impact costs $(\lambda_L/2)r^2$ per day in excess of the prevailing price; a constant rate $r = x/d$ is optimal under a convex per-day cost, so the total accumulation cost over the window is

Lemma 1 (R1: Accumulation cost from a dealer participation cap). *Under linear dealer price impact $\lambda_L(L)$ and a constant-rate purchase of x over d days,*

$$C(x; d, L) = \frac{\lambda_L(L)}{2} \frac{x^2}{d}, \quad (1)$$

and therefore, for $x > 0$ and $d > 0$,

$$C_x = \frac{\lambda_L x}{d} > 0, \quad C_{xx} = \frac{\lambda_L}{d} > 0, \quad C_{xd} = -\frac{\lambda_L x}{d^2} < 0, \quad (2)$$

with the exact horizon scaling $C(x; \kappa d, L) = \kappa^{-1} C(x; d, L)$ for any $\kappa > 0$.

Derivation sketch. Per-day excess cost at rate r is $\int_0^r \lambda_L u \, du = (\lambda_L/2)r^2$. Minimizing $\sum_{t=1}^d (\lambda_L/2)r_t^2$ subject to $\sum_t r_t = x$ is a convex problem with symmetric constraint, so $r_t = x/d$; substituting gives $d \cdot (\lambda_L/2)(x/d)^2 = (\lambda_L/2)x^2/d$. The derivatives and the degree -1 homogeneity in d follow by differentiation. $\square$

Three consequences matter. First, the cross-partial $C_{xd} < 0$ — the object on which the paper’s only theoretical novelty rests — is now a *theorem*, not an assumption. Second, the cost is monotone in liquidity through $\lambda_L(L)$, and the liquidity object in the model is the same object the regressions use. Third, the horizon scaling is quantitative: the reform’s move from $d \approx 7$ to $d = 5$ business days raises the quadratic component of accumulation cost, at any fixed post-trigger increment, by the factor $7/5 = 1.4$, i.e. by 40%. The same mechanism delivers, for free, an endogenous pre-filing run-up whose magnitude scales with $\lambda_L x/d$ — a sharper cross-sectional prediction about the split between leakage and filing CAR than a reduced-form cost can generate.

What this shortcuts. Equation (1) is a static reduced form for a dynamic problem. A full deadline-Kyle treatment in the manner of Collin-Dufresne and Fos (2016) would endogenize the intensity path, let the activist trade off impact against information leakage over the window, and generate a non-constant rate near the deadline; Back et al. (2018) is the corresponding strategic benchmark with an endogenous informed trader. What that machinery would add is the *shape* of the within-window trading path, not the sign of C_{xd} or the $1/d$ scaling, both of which survive as long as per-period impact is convex. Collin-Dufresne and Fos (2015) establish that informed trading inside the 13D window is a real and measurable phenomenon, which is what licenses treating the window as a binding trading horizon at all; Kyle (1985) supplies the linear-impact primitive.

4.3 The deadline and who it binds

The participation cap converts the legal window into a feasibility constraint. Over the d post-trigger days the activist can add at most $\phi \text{ADV}_i d$ shares, i.e. an increment

$$\bar{x}_i(d) = \frac{\phi \text{ADV}_i d}{S_{\text{out},i}}. \quad (3)$$

Define the target’s *required trading days* per five percentage points of ownership at the participation cap,

$$\text{RTD}_i \equiv \frac{0.05 \times S_{\text{out},i}}{\phi \times \text{ADV}_i}, \quad \text{so that} \quad \bar{x}_i(d) = 0.05 \frac{d}{\text{RTD}_i}. \quad (4)$$

Proposition 1 (R2: Corner-solution heterogeneity and the derived exposure measure). *Let $x^u(d)$ denote the unconstrained optimum of the activist’s problem $\max_x b(x, d) - C(x; d, L) - F$, where x is the post-trigger increment (the untimed toehold is fixed at the threshold), b is the gross benefit of a stake and F a fixed campaign cost. Then the activist’s increment is $x_i^*(d) = \min\{x^u(d), \bar{x}_i(d)\}$, and:*

- (i) $\bar{x}_i(d) \geq 0.05 \iff \text{RTD}_i \leq d$: the window permits an undisclosed post-trigger increment of at least another five percentage points — a doubling of the threshold stake — if and only if $\text{RTD}_i \leq d$. The maximum stake acquirable before disclosure, $0.05 + \bar{x}_i(d)$, is decreasing in RTD_i .
- (ii) Under the five-percentage-point benchmark increment, the constrained set is $\{i : \text{RTD}_i > d\}$ (in general, $\{i : \bar{x}_i(d) < x_i^u(d)\}$); it is increasing (in the set-inclusion sense) as d falls, and the shadow value of the constraint, $b_x(\bar{x}_i(d), d) - C_x(\bar{x}_i(d); d, L) \geq 0$, is increasing in RTD_i on that set.
- (iii) Hence treatment intensity is a model object: the reform's dose at target i is a monotone function of RTD_i/d , and the binary treated indicator is $\mathbf{1}\{\text{RTD}_i > d\}$.

Part (i) is the model's reading of the accumulation ladder tabulated in the adopting release (Securities and Commission 2023): the large majority of filers who complete accumulation within five business days of the trigger, and the near-universal share who complete within ten, sit where the window's capacity exceeds their desired increment — the deadline is slack for them. That ladder is therefore not an empirical caveat bolted onto the design; it is the model's own prediction about which targets the reform can possibly bind on. Part (iii) is the point of the whole subsection: the empirical exposure variable is *derived* from (3)–(4), uses only pre-trigger, target-level, price-free inputs, and is defined for one-time filers — which is what makes activist $\times$ quarter fixed effects feasible in the empirical section.

Proposition 2 (R3: The window comparative static). *Suppose $b(\cdot, d)$ is C^2 and the objective is strictly concave in x ($b_{xx} - C_{xx} < 0$). Then on the interior (unconstrained) region*

$$\frac{dx^*}{dd} = -\frac{b_{xd} - C_{xd}}{b_{xx} - C_{xx}} > 0 \iff b_{xd} > C_{xd} = -\frac{\lambda_L x^*}{d^2}. \quad (5)$$

On the constrained region $x_i^* = \bar{x}_i(d)$ and $dx_i^*/dd = \phi \text{ADV}_i/S_{\text{out},i} > 0$ unconditionally.

Proof sketch. Monotone comparative statics on a concave objective: the sign of dx^*/dd is the sign of the cross-partial of the objective, $b_{xd} - C_{xd}$, by the implicit function theorem applied to the first-order condition, the second-order condition supplying the negative denominator. The constrained case is immediate from (3). $\square$

Honesty condition (SC). Condition (5) is not automatic and we flag it as the model's load-bearing single-crossing requirement. The benefit b depends on d through the entry margin: a longer window means a later filing, hence later certification and a possibly lower probability that a bid arrives while the activist is still positioned, so b_{xd} may be negative. The condition states that the marginal benefit of stake must not fall too fast in d — specifically, $|b_{xd}| < \lambda_L x^*/d^2$ whenever $b_{xd} < 0$. It is a genuine restriction, it is weakest exactly where the accumulation friction is largest (high λ_L , low d , high x — the constrained targets), and it holds vacuously on the corner. We state it as a condition rather than assume it away.

4.4 Bidder entry: certification against deterrence

Potential acquirers are ex ante identical and free to enter: each pays a search and evaluation cost s to learn its own valuation, and enters until expected profit is zero. Two forces run in opposite directions, and the paper's earlier companion model carried only the second.

Certification. A 13D filing is a public, costly-to-fake signal that a large, sophisticated holder has identified the target and will not defend the incumbent management, which lowers s : the filing does part of the acquirer’s screening for it. This is the solicitation-and-search technology of Corum and Levit (2019) and the activist-as-intermediary/brokerage technology of Burkart and Lee (2022); we take both as given rather than rebuild them.

Deterrence. Disclosure raises the price the acquirer expects to pay: the disclosed stake x and the engagement it implies are capitalized into P^F , and a larger, better-informed bloc bargains harder. This is the preemption channel of Fishman (1988) and the empirical toehold/disclosure-reform finding of Bonetti et al. (2020), and it is the force that produced $\partial p/\partial \pi < 0$ in the author’s companion model. That result is nested here, not renounced: it is the ε_{det} term below.

Let $h(s, B^e)$ be the free-entry bid hazard, decreasing in both arguments, and let e denote the *earliness* of disclosure (minus the filing lag), holding the campaign fixed.

Proposition 3 (R4: Certification versus deterrence). *Define the two nonnegative elasticities*

$$\varepsilon_{\text{cert}} \equiv \left(-\frac{\partial \log h}{\partial \log s} \right) \left(-\frac{\partial \log s}{\partial e} \right), \quad \varepsilon_{\text{det}} \equiv \left(-\frac{\partial \log h}{\partial \log B^e} \right) \left(\frac{\partial \log B^e}{\partial e} \right).$$

Then $\frac{d \log h}{de} = \varepsilon_{\text{cert}} - \varepsilon_{\text{det}}$, so the bid hazard rises after earlier disclosure if and only if $\varepsilon_{\text{cert}} > \varepsilon_{\text{det}}$.

The sign is therefore an empirical question with named heterogeneity. Certification should dominate where search is expensive and the counterparty set is not obvious: opaque or hard-to-value assets, few natural strategic buyers, cold industry M&A markets, and Item 4 language expressing sale or strategic-alternatives intent. Deterrence should dominate where acquirers are already searching — hot industry M&A intensity, an obvious short list of strategic buyers — so that the certification elasticity is small while the price effect is not. One complication is worth stating now: the reform does not move e alone. Through R3 it also lowers x^* , which lowers B^e and therefore *raises* entry, a third term partially offsetting deterrence. The empirical design must not attribute the composite to certification.

4.5 Bargaining and appropriability

If a bid arrives, the final offer splits the surplus between the acquirer’s standalone valuation V_a and the improved value Y under a successful campaign:

$$B = V_a + \beta(x) (Y - V_a), \quad \beta(x) = \bar{\beta} \lambda_{\text{app}}, \quad \lambda_{\text{app}} \in [0, 1]. \quad (6)$$

The discipline on β is the point. An unrestricted increasing $\beta(x)$ would assume away the free-rider structure of Grossman and Hart (1980b), which is exactly the circularity the companion paper’s appendix was written to remove. We therefore import that appendix’s *conclusion* and none of its primitives: $\lambda_{\text{app}} = 1 - q(1 - \gamma)\psi$, where q is the probability that a fringe raider appears at the disagreement node, $\gamma \in [0, 1]$ is the portability of the activist’s improvement under a different owner, and $\psi \in [0, 1]$ is a pivotality factor. Appropriability falls with the fringe-raid probability and rises with portability. Two remarks port directly and become the cross-sectional heterogeneity table of the empirical section: (a) the primitives are *indexable* — q by fringe M&A intensity in the target’s industry-year, γ by the strategic-versus-financial composition of likely acquirers and by asset redeployability, the charter dilution parameter by state and charter antitakeover provisions, and pivotality by the bloc’s stake against control thresholds; and (b) λ_{app} *jumps up discontinuously* as the bloc’s stake crosses the blocking threshold $1 - \tau_c$,

a testable stake-size threshold effect that is directly about x and therefore directly about the deadline. We do *not* import the tender game itself: the entry-cost distribution, the synergy distribution, charter dilution and the control threshold are four new primitives a fourteen-page model cannot afford. (Note the symbol clash: charter dilution in that appendix was ϕ ; here ϕ is the participation cap. The paper renames the former ϕ_c .)

4.6 Measurement lemma

Total target gain from the unaffected price to the final offer decomposes exactly:

$$G_{\text{tot}} \equiv \log \frac{B}{P^0} = \underbrace{\log \frac{P^F}{P^0}}_{G_{\text{fil}}} + \underbrace{\log \frac{P^B}{P^F}}_{G_{\text{run}}} + \underbrace{\log \frac{B}{P^B}}_{G_{\text{mkp}}}. \quad (7)$$

This is an identity, and we label it as one. Its non-trivial content is the following.

Lemma 2 (R5: Markup floor and the invariance restriction). *Let the bid-eve price be the discounted expectation of resolution, $P^B = \delta [\varrho B + (1 - \varrho)P^{\text{fail}}]$, where $\varrho \in (0, 1]$ is the completion probability and $P^{\text{fail}} < B$ the freestanding value. Then:*

- (a) **Floor.** $G_{\text{mkp}} = \log(1/\delta) - \log(\varrho + (1 - \varrho)P^{\text{fail}}/B) \geq \log(1/\delta) > 0$, i.e. in simple returns the measured markup is bounded below by $(1 - \delta)/\delta$ plus a strictly positive deal-risk term. A fully anticipated deal still shows a positive markup; a shrinking measured markup is therefore not, by itself, evidence of anything.
- (b) **Invariance.** Holding the transaction fixed (B and P^0 unchanged), (7) implies $dG_{\text{mkp}} = -(dG_{\text{fil}} + dG_{\text{run}})$.

Part (b) is the only thing in the neighbourhood that distinguishes capitalization from value destruction in data: under pure capitalization the three components reshuffle and G_{tot} is invariant; under value destruction G_{tot} falls. Celentano and Levine (2025) assert the capitalization reading for their structural counterfactual; Betton et al. (2014) document run-up/markup substitution once run-ups are value-adjusted — Schwert (1996) found little substitution without that adjustment, and the difference is precisely the invariance test. Our lemma states the exact testable content of the claim, and the empirical section implements it in *event-time-invariant* form, because the naive calendar-window version carries a truncation artifact running opposite to its own prediction.

4.7 The policy sufficient statistic

Result 1 (R6: Welfare derivative for target shareholders). *Let $W(d)$ be expected target-shareholder gain. On the interior region the envelope theorem kills the indirect effect of d through x^* on the activist's own payoff, so every dx^*/dd term surviving in $W'(d)$ is a pure externality; on the constrained set the window's shadow value enters the activist's own payoff first-order, with the constrained response $dx^*/dd = \phi \text{ADV}_i / S_{\text{out},i}$ itself part of term (iii)'s comparative static. In both regions,*

$$\frac{dW}{dd} = \underbrace{\frac{\partial \mathbb{P}(\text{bid})}{\partial d} \mathbb{E}[G_{\text{tot}} \mid \text{marginal deal}]}_{(i) \text{ entry margin}} + \underbrace{\frac{\partial \mathbb{P}(\text{camp})}{\partial d} \mathbb{E}[G_{\text{tot}} \mid \text{marginal campaign}]}_{(ii) \text{ campaign-supply margin}} + \underbrace{\mathbb{P}(\text{bid}) \frac{\partial \mathbb{E}[G_{\text{mkp}}]}{\partial \beta} \beta'(x^*) \frac{dx^*}{dd}}_{(iii) \text{ stake-bargaining margin}} \quad (8)$$

Term (i) has the sign of R4 and is therefore an estimand, not an assumption. Term (ii) is nonnegative — a longer window supports more campaigns — so it enters the reform’s welfare change ($\Delta d < 0$) with a minus sign: this is the chilling term, and it is the paper’s headline object. Term (iii) is the channel T2 flagged as omitted and it runs through R3 and §4.5: a shorter window lowers x^* , which lowers $\beta(x^*)$, which lowers the conditional markup independently of both other margins.

The measurability claim needs one further step, and it is the step that makes the formula usable rather than aspirational. Both conditional expectations in (8) are *marginal*, not average, objects — marginal deals are by construction absent from the counterfactual regime, and deterred campaigns are unobserved. Substituting the average observed premium would be exactly the LATE-versus-ATE substitution the identification section rejects elsewhere. The resolution is that the reform’s *compliers are the marginal units*: an intensity design identified off RTD_i recovers the behaviour of campaigns at the margin of being deterred. The exposure-based estimands are therefore the objects the welfare weights require, and the model and the regressions are estimating the same thing. This also disciplines the accounting: because accumulation moves the price, part of G_{fil} is a transfer from selling shareholders to the activist rather than a gain, and the welfare object in (8) is defined over a fixed shareholder cohort dated at trigger-minus- k , with the transfer term separated explicitly.

4.8 What version 1 deliberately omits

The signalling block. P^F is treated as a Bayes update against an *exogenously garbled* filing: nature adds noise to the disclosed $(x, \text{purpose})$ pair, and misstatement of Item 4 purpose carries an exogenous securities-liability cost. We do not solve the game in which an informed activist chooses x as a signal. The reason is worth stating rather than hiding, because a referee will find it either way. A continuous signal chosen by a privately informed type admits pooling, partial-pooling and separating equilibria; without a stated refinement and a uniqueness claim, P^F is not a well-defined object and the mapping from the filing CAR to a model quantity is not identified. Worse, there is a genuine dilemma at the closure point: if B in (6) is independent of P^F , the capitalization proposition is a tautology (the denominator moves and the numerator cannot); if B depends on P^F through a market-price floor on offers, then $P^F \rightarrow B \rightarrow P^F$ is a fixed point requiring a contraction condition. We choose the first horn *explicitly*, which is precisely why R5 is labelled a measurement lemma and why capitalization is not this paper’s headline. Endogenizing the filing price is the natural follow-on paper, not a footnote in this one.

Not modelled, by design. The operational-versus-sale split as a second type dimension (§4.1); the tender game underlying λ_{app} (§4.5); bidder competition and repeated fringe arrival; the within-window trading path (§4.2); and cash-settled derivative positions, which the empirical Item 6 substitution test treats as an outside option without pricing it. This last omission is the one we are least comfortable with, since the substitution margin is a headline empirical result; the model’s role there is to supply the constrained set $\{\text{RTD}_i > d\}$ that predicts *who* substitutes, not the price at which they do.

Mechanism budget. The specification above is the neutralized one: a scalar type, a quadratic accumulation cost, an exogenously garbled filing, a free-entry bidder cutoff, and a bargaining share restricted by a single appropriability parameter. Ten to fourteen pages is feasible for *this* model and is not feasible for the version with a two-dimensional type, an endogenous informative

stake, endogenous purpose, endogenous entry, endogenous β , and three endogenous prices — that specification is eighteen to twenty-two pages plus appendix, which is how the companion draft reached its current length. We therefore adopt a standing rule: *any mechanism added after the model is drafted requires deleting another*.

5 Empirical Design and Data Acquisition

This design implements the identification matrix and data-feasibility findings of the E1 and E2 memos rather than the pivot memo’s original activist-level designs. The headline test is the chilling/substitution result: entry and composition dose-response on a predetermined target-level exposure measure, paired with Item 6 cash-settled-derivative substitution through the hole the Commission left open when it declined to adopt proposed Rule 13d-3(e). The event-time-invariant anatomy of activist target gains is the second act. Capitalization is a measurement lemma inside that anatomy, not a headline: Celentano and Levine (2025) own the capitalization interpretation and Betton et al. (2014) own the substitution evidence this design adjudicates against realized post-reform data.

5.1 Setting and Identification Stance

The reform is a single adoption event. Rule 13d-1(a) was amended to shorten the initial Schedule 13D filing deadline from ten calendar days to five business days, effective February 5, 2024 (Securities and Commission 2023). There are no never-treated units: every initial 13D filer is subject to the same deadline after that date, so identification runs entirely through continuous treatment intensity rather than a treated/control contrast. Let x denote the activist’s eventual stake (fraction of shares outstanding), L target liquidity (average daily volume over the six months ending 60 trading days before the trigger, or its Amihud illiquidity companion), ϕ the maximum share of L tradable per day without alerting the market, and d the legal filing window in business days ($d \approx 7$ pre-reform, $d = 5$ post-reform). Define the exposure measure

$$\text{RTD}_i = \frac{0.05 \times S_{\text{out},i}}{\phi \times \text{ADV}_i},$$

with $S_{\text{out},i}$ shares outstanding: the number of trading days required to move five percentage points of ownership at the participation cap. The statutory clock starts when the activist crosses the 5% threshold; over the following d business days at most $\phi \times \text{ADV}_i \times d$ shares can be added, so the window permits an undisclosed increment of at least another five percentage points iff $\text{RTD}_i \leq d$. The binary exposure is $\mathbf{1}\{\text{RTD}_i > d\}$ and the continuous dose is RTD_i/d , the five-percentage-point increment being the stated normalisation. Because RTD_i is built from pre-trigger, predetermined target characteristics, it is defined for every filing, including one-time filers, and does not depend on the campaign’s realized outcome. Because the untimed pre-trigger phase can extend into any finite measurement window, the ADV window is lagged (six months ending 60 trading days before the trigger), residual own-trading in it is diagnosed from the Item 5(c) sixty-day transaction tables, and every specification is re-estimated with a 120-trading-day lag — extending the caution of Collin-Dufresne and Fos (2015) from controls to the exposure measure itself.

Assumption 1 (Continuous-dose parallel trends). *In the absence of the reform, outcomes for targets at every level of RTD_i would have evolved along parallel paths, and the dose-response function is stable across the exposure distribution over the estimation window.*

Under this assumption a single interaction coefficient from

$$Y_{i,q} = \alpha_{a(i),q} + \gamma_{\ell(i),q} + \beta \text{Post}_q \times \text{RTD}_i + \varepsilon_{i,q}$$

identifies an average causal response only if the dose-response is homogeneous; if it is not, β is a weighted average of heterogeneous responses whose weights need not be well-behaved. We therefore treat β as a summary statistic only and report the primary result nonparametrically,

$$Y_{i,q} = \alpha_{a(i),q} + \gamma_{\ell(i),q} + \sum_k \beta_k \mathbf{1}[\text{RTD-bin}_i = k] \times \text{Post}_q + \varepsilon_{i,q},$$

across exposure bins, with $\alpha_{a(i),q}$ an activist $\times$ quarter fixed effect and $\gamma_{\ell(i),q}$ a liquidity-decile $\times$ quarter fixed effect (subsection 2).

Anticipation and the double donut. Anticipation predates the effective date by two years: the amendments were proposed and published at 87 FR 13846 (Mar. 10, 2022), the comment period was reopened Apr. 28, 2023, and the release was adopted (dated) October 10, 2023, four months before the February 5, 2024 effective date (Securities and Commission 2023). We donut out the October 2023–February 2024 window from any pre-period, and separately test robustness to donuts around the March 2022 proposal and the April 2023 reopening, since either could have partially treated behavior ahead of the effective date.

Bundling defense. Release 33-11253 moved four things at once: it shortened the initial 13D deadline to five business days; it shortened Schedule 13D amendments to two business days; it restructured every Schedule 13G filing and amendment deadline; and it mandated XML-based structured filing for Schedules 13D and 13G, with compliance not required until December 18, 2024 (Securities and Commission 2023). Separately, and centrally to the substitution channel, the release declined to adopt proposed Rule 13d-3(e) on cash-settled derivative securities, instead amending Item 6 to clarify (without expanding) disclosure of derivative interests (Securities and Commission 2023). The two-business-day amendment rule directly moves "post-filing accumulation" *independent of the initial-deadline change*, so that outcome is dropped as a first-stage margin. We restrict tier-1 outcomes (subsection 3) to those only the initial-deadline change can move: trigger-to-filing accumulation share, stake at filing, entry counts and composition, and Item 6 derivative-language rates. As a composition test for the simultaneous 13G restructuring, we tabulate the switching rate of near-5% holders between Schedule 13G and Schedule 13D filings in the quarters surrounding the reform; a level shift in that rate is evidence the 13G bundling, not the 13D deadline, is doing the work on any outcome defined over the pooled 13D/13G population.

5.2 Exposure: From Activist History to a Predetermined Target-Level Measure

The pivot's original exposure variable was a leave-one-out mean of the activist's own historical accumulation behavior, with activist fixed effects γ_j absorbing the between-activist variation the measure needs to identify anything. That design is infeasible at the relevant population. Polk et al. (2024) report 9,685 initial Schedule 13D filings by 5,863 reporting entities over 2001–2022, or 1.65 filings per entity. This project's own 300-filing sample tells the same story directly: 259 unique filer CIKs, 29 filers with two or more filings, and exactly 3 filers with three or more.

A leave-one-out mean needs at least two pre-2024 campaigns per activist, and γ_j additionally restricts identification to activists observed both before and after February 2024 – on the order of a few dozen funds. External validity then narrows from "13D filers" to "Sharkwatch-50-type funds," precisely the population for which the accumulation constraint binds least, since these are the filers best able to pre-position and substitute into derivatives.

We therefore adopt RTD_i (subsection 1) as the primary exposure measure and retain activist-history exposure only as a secondary heterogeneity cut (an indicator for activists observed in at least two pre-2024 non-corporate-action campaigns), not as the identifying variable. Because RTD_i is defined at the target level, it makes activist $\times$ quarter fixed effects estimable: identification runs within an activist across targets and quarters, which absorbs activist-level shocks that a target-industry $\times$ time fixed effect cannot reach – fund flows, a manager’s strategy shift, or the differential ramp-up of universal-proxy-card-driven board-contest returns across 2023–2024.

Covariate imbalance and the FE response. Realized accumulation behavior is not a clean proxy for the legal constraint: the Commission’s own Table 6 shows that non-corporate-action campaigns under 90% accumulated by the amended deadline differ sharply from campaigns that finished on time. Average target market capitalization is \$1.8B versus \$916M; the Amihud illiquidity ratio is 0.09 versus 0.13 (lower Amihud is higher liquidity, so the campaigns that did *not* finish by the deadline sit in the *more* liquid, and much larger, targets: realised slow accumulation marks big campaigns, not illiquid stocks); the prominent-activist share is 43.6% versus 29.8%; the filing-window cumulative abnormal return is 17.2% versus 5.7%; and the average increase in shareholder value per campaign is \$222M versus \$36M (Securities and Commission 2023). RTD_i , built from liquidity and size rather than realized accumulation, is designed to avoid mechanically encoding "large campaign by a large activist," but it will still correlate with target size and illiquidity by construction. We therefore saturate every specification with liquidity-decile $\times$ quarter and size-decile $\times$ quarter fixed effects in addition to activist $\times$ quarter, and report the covariate balance of targets across RTD_i bins as a standing diagnostic table before any point estimate is presented, following the logic of stealth-trading detection-avoidance constraints on accumulation speed (Collin-Dufresne and Fos 2015; Collin-Dufresne and Fos 2016) and the liquidity-governance tradeoff that motivates using L as a covariate rather than an outcome (Edmans et al. 2013).

5.3 Outcome Hierarchy and the MDE-Honesty Table

Before any regression is run, we state the population magnitude the design can hope to detect. By the Commission’s own Table 3, the accumulation window binds materially (under 90% completed by the amended deadline) on 3% of non-corporate-action campaigns, about 7 per year; only 1% fall under 75% completed, about 1 per year (Securities and Commission 2023). Table 6 row 8 shows that the 20% of campaigns that continue accumulating past the deadline acquire, on average, only 5.9% of their reported stake inside the window (Securities and Commission 2023). Combined, the population-average treatment intensity of the accumulation channel is on the order of a tenth of a percentage point of shares outstanding. No result below is credited as more than this magnitude can support.

Table 2 sets the outcome hierarchy. Tier 1 outcomes are near-term, extensive-margin or predetermined, and powered at the realistic post-reform sample size (roughly 540–700 non-corporate-action campaigns over 2024–2026, before any repeat-activist restriction). Tier 2 outcomes require price data and are moderately powered. Tier 3 is reported descriptively, with confidence intervals, and is never presented as a headline test: the Post $\times$ RTD $\times$ SaleIntent

triple interaction on bid hazard has an ex-ante minimum detectable effect of 17–31 percentage points on a roughly 17-percentage-point base rate, and the materially-exposed post-reform cell (the SEC’s own under-90% margin) contains on the order of three bids over the sample window. A test that cannot distinguish a null from a doubling of the base rate is not a test.

Table 2: Outcome hierarchy and MDE honesty

Tier	Outcome	Level	Power / status
1 (head-line)	Entry counts & composition	liquidity-decile $\times$ quarter	extensive margin, unselected by conditional-on-filing treatment; best powered
1	Stake at filing	campaign	powered; predicted to fall under a deterrence channel
1	Trigger-to-filing accumulation share	campaign	powered; only the initial-deadline change can move it
1	Item 6 derivative-language rate	campaign	powered; the substitution channel through the dropped 13d-3(e) hole
1	Filing timing (delay)	campaign	mechanical – reported, not credited as behavioral evidence
2	Filing-date $[-1, +1]$ CAR, share of total gain	campaign	event-time-invariant; moderately powered, requires CRSP
2	Pre-filing informed-bystander drift	campaign	moderately powered, requires CRSP
3 (descriptive)	6- and 12-month bid hazard	campaign	CIs only; triple interaction excluded (MDE 17–31pp on a ~ 17 pp base; ~ 3 bids in the materially-exposed post-reform cell)

5.4 First-Stage Battery: EDGAR-Only, Price-Free, Pre-Registered

Before any model investment, we run a first-stage battery that requires no price data and is pre-registered against ex-ante minimum detectable effects computed from the Commission’s own Tables 3, 5, and 6, not from the realized sample. Five components:

1. **Untrimmed delay distributions, split by corporate-action status.** This project’s own fact1 pipeline reports a post-reform (2024Q3–Q4) median delay of 5.0 business days versus 7.0 pre-reform (2023Q2–Q3), and a within-5-business-day share of 75.6% post versus 35.7% pre. The 75.6% figure depends on an undocumented trim: it drops observations above 60 business days, and recomputing directly from the full set of 96 non-missing post-period delay observations in this project’s own filing-level data gives a within-5-business-day share of 70.8% (68 of 96), not 75.6%. The pooled sample also mixes corporate-action and non-corporate-action filings, which the Commission finds have very different late-filing rates (34% versus 11%) (Securities and Commission 2023). Both defects are fixed here: we report the untrimmed compression statistic with the trim as a footnoted robustness check,

and we split every delay statistic by tabular-Item-5(c)-history status once that classifier exists (subsection 7).

2. **Stake-at-filing distributions**, pre/post, to test the deterrence prediction directly on a predetermined outcome.
3. **Entry and composition counts by a coarse, price-free RTD proxy** (e.g. pre-trigger share turnover deciles), since the full RTD_i measure requires ADV data whose access is gated (subsection 7); this proxy lets the first-stage pass run before that gate resolves.
4. **Item 6 derivative-language rates**, pre/post, as the direct test of substitution through the derivative-disclosure hole (Securities and Commission 2023).
5. **13G-to-13D switching**, as the composition test described in subsection 1.

The gate rule: filing-delay compression and post-deadline pre-filing accumulation are mechanical identities under the reform and cannot be used to pass any decision gate, because a merely compliant world produces both with zero behavioral content.

5.5 Inference Battery

With one adoption date, continuous dose, and no staggered-timing structure, the applicable threat is not negative-weighting in a staggered design but a fragile continuous-dose estimate that could be an artifact of the specific placebo date, the specific exposure assignment, or the 2023–2025 M&A and rate cycle. The battery:

- at least 500 placebo reform dates drawn from 2015–2023, re-estimating the full specification at each and reporting the realized estimate’s percentile in the placebo distribution;
- at least 500 permutations of RTD_i across activists, holding the panel structure fixed, for the same percentile test;
- exposure-bin quarterly event studies from 2019 onward, with a joint pre-trend F-test as a precondition for any causal claim;
- a non-US activist-block sample (UK DTR5 / EU Transparency Directive notifications) over the same calendar window, used strictly as a control for the global M&A and rate cycle and never as a legal-threshold comparison, since the disclosure regimes are not otherwise comparable.

5.6 Anatomy of Target Gains (Second Act)

The second act decomposes total target gains into filing, run-up, and markup components: $G_{\text{fil}} = \log(P^F/P^0)$, $G_{\text{run}} = \log(P^B/P^F)$, and $G_{\text{mkp}} = \log(B/P^B)$, where P^0 is the unaffected price, P^F the post-filing price, P^B the bid-eve price, and B the final offer. We fix P^0 at a common horizon of 60 trading days before the trigger date in both the pre- and post-reform cohorts (robustness at 30 and 120 trading days), so that the anchor is not itself shifted by treatment. Because the pre-trigger toehold phase is untimed, the activist’s own buying can reach into the anchor window — mechanically so where $RTD_i > 60$ — so we diagnose pre-trigger own-trading in the anchor window from the Item 5(c) sixty-day transaction tables and lean on the 120-day anchor where contamination is flagged.

The reform compresses the trigger-to-filing interval by roughly two business days in this project’s own data (median delay 7.0 to 5.0 business days), which mechanically shrinks the calendar span inside G_{fil} and lengthens the span inside G_{run} with zero behavioral content – and does so in the direction opposite the reform’s naive prediction that earlier disclosure should raise G_{fil} ’s share of total gain. Every prediction here is therefore restated on event-time-invariant objects: the filing-date $[-1, +1]$ cumulative abnormal return as a share of total gain, with the trigger-to-bid horizon held fixed, rather than on calendar-spanning segments. Two placebos separate the mechanical artifact from any behavioral effect. The *mechanical placebo* recuts pre-reform campaigns as if they had filed under a five-business-day rule, which isolates the pure truncation artifact directly from pre-period data. The *behavioral placebo* restricts to already-fast filers – campaigns that filed within five business days even under the old ten-calendar-day rule – for whom the reform changes nothing mechanically, so any measured shift in their gain decomposition is a pure cycle or composition effect, not a deadline effect.

Where an M&A outcomes database is unavailable (subsection 7), the acquired-target subsample is hand-collected from 8-K and DEFM14A filings, following measurement conventions from the M&A literature on toeholds, premia, and deal outcomes (Betton et al. 2014; Eckbo et al. 2025). Capitalization – completion probability up, conditional premium down through weakened target resistance – is treated throughout as a measurement lemma inside this anatomy, with the interpretation credited to Celentano and Levine (2025) and the resistance-weakening mechanism to Corum and Levit (2019), not restated as this paper’s own headline finding. Heterogeneity is explored through the appropriability remnants of the disclosure-timing channel: industry M&A intensity as a proxy for fringe-raid probability, strategic-buyer share as a proxy for the portability of the activist’s information rents to a bidder (the broker/intermediary framing of Burkart and Lee (2022)), and a regression-discontinuity-style check around the 5% stake threshold itself. The classic activist-return decomposition of Brav et al. (2008) anchors the overall magnitude of gains against which this design’s event-time-invariant restatement is compared.

5.7 Data-Acquisition Map

Table 3 inventories what exists in this project’s pipeline against what the design above requires, following the E2 memo’s build inventory. Two hard access gates dominate the schedule: CRSP/WRDS access is unconfirmed, and no M&A outcomes database is currently held. Both are resolved at week 1–2, not discovered at week 7. Merger-background chronology extraction is excluded from the critical path entirely – no chronology code, corpus, or design note exists anywhere in this repository, so any claimed comparative advantage there does not carry into this design.

Sample. The core estimation sample is 2022–2026, bracketing the reform symmetrically while limiting exposure to the Item 5(c) extraction backlog; the Commission’s own published 2011–2021 tables serve as a pre-period anchor without requiring this project to re-extract that history. A 2010–2021 backfill is undertaken only after Item 5(c) extraction quality is proven at scale on the core sample.

5.8 Gates, Cut Order, and Scoop Defense

Gate 0 (week 1): advisor ratification. Gate 1 (weeks 1–2): access. CRSP/WRDS and the Bloomberg M&A checklist are resolved as a hard go/no-go, since every price-based tier-

Table 3: Data-acquisition map

Component	Exists now	Build (wks, solo part-time)	est. solo	Gate	Blocking risk
13D + amendment enumeration	partial	1–2		none (running today)	low–medium
Item 5(c) transaction-table extraction	none	4–8+		pre-model	high; SEC itself used programmatic extraction plus significant manual review of heterogeneous tables
Item 4 purpose classification + blinded validation	none	3–5		95% auto-vs-manual agreement on the 200-filing validated subset (not the raw universe)	medium–high
Trigger-date parser fix + hand-verification against Item 5(c) tables	partial (parse rate 0.64–0.68)	2–3		pre-model	medium
Post-Dec-2024 XML ingestion	partial	1–2		none	low mechanically; window only ~18 months, cannot carry the sample alone
CRSP/WRDS access	unconfirmed	1–2 once access lands		week-1 go/no-go	gates price legs; EDGAR-only fallback per cut order
Amihud / ADV construction	none	1–2 after CRSP		gated on CRSP/WRDS	high
M&A outcomes	none	4–8 (EDGAR 8-K/DEFM14A route)		week-2 source decision (Bloomberg checklist vs. EDGAR route)	high
Merger-background chronologies	none	excluded from critical path		appendix / second paper only	n/a – descoped

2/tier-3 outcome and the anatomy of gains depend on one or the other and neither has been confirmed to date. **Gate 2 (weeks 2–4): pre-registered EDGAR-only first-stage pass**, using the battery in subsection 4 and the non-mechanical margin rule stated there. If at least one non-mechanical margin moves at its pre-registered MDE, the deadline- accumulation model receives further investment (beyond the frozen provisional lens of Section 4) as the disciplining lens for the headline result; if none moves, the paper is re-aimed at the accounting lemma, the corner-solution model, and a well-measured policy null, with the anatomy of gains carrying the paper rather than the chilling result.

Published cut order if the schedule slips: merger chronologies (already excluded) → the signaling block → the full deal-universe anatomy (shrinking to the hand-collected acquired-target subsample) → the 2010–2021 backfill → an EDGAR-only fallback carrying only the first-stage facts and the exposure-composition results.

Scoop defense. The first-stage facts and the exposure design are posted to SSRN once the exposure measure locks, at approximately month 4–5, ahead of the full anatomy-of-gains work. The novelty sweep and the publication status of Celentano and Levine (2025) are re-checked at every subsequent revision, since that paper’s own capitalization interpretation is the one this design’s measurement lemma must continue to cite rather than re-derive.

6 What We Expect to Learn

The empirical program has two headline margins — behavioural response to the acceleration (entry and composition) and substitution into cash-settled derivatives — and each can turn out responsive or null, giving four worlds. Each world teaches something different, and each has a known destination.

First, the *responsive-with-substitution* world: campaign entry falls or tilts towards liquid targets in proportion to predetermined required-trading-days exposure, and Item 6 derivative usage rises in exactly the high-exposure targets where the five-business-day deadline binds. This is the intended headline: the acceleration did not shrink activist stakes so much as move them into instruments the final rule chose not to cover, a regulatory-arbitrage result with direct policy teeth, adjudicating the \$810 million per year projection of Securities and Commission (2023) against realised behaviour. This world supports the upper tier of outcomes.

Second, the *responsive-without-substitution* world: entry and composition respond to exposure but derivative usage stays flat. Here the reform genuinely changed who campaigns and where, the chilling projection is validated in direction though perhaps not in magnitude, and the paper’s contribution is the first realised measurement of how a disclosure deadline reallocates activist attention across the liquidity distribution.

Third, the *null-with-substitution* world: entry is unmoved (consistent with the SEC’s own tables showing most filers had already completed accumulation within five business days) but derivatives rise in exposed targets. This says the constraint was never the campaign but the form of ownership, and it isolates the dropped Rule 13d-3(e) as the operative policy choice rather than the headline deadline change.

Fourth, the *null-everywhere* world: activists simply comply on every margin. This is the modal ex ante outcome given the regulator’s compliance statistics, and it is still a publishable policy evaluation: a pre-registered, well-powered null on a contested rule, with honest minimum detectable effects, adjudicating a large money projection with the realised data the regulator itself has not revisited. That world lands at the honest tier (JCF/RF), and we regard the

chilling/substitution worlds as the upside (RFS and above), not the promise.

In every world the second act stands: the event-time-invariant anatomy of activist target gains, decomposing total returns into the filing return G_{fil} , the post-filing run-up G_{run} , and the bid markup G_{mkp} , shows where takeover value reaches target shareholders and disciplines the capitalisation measurement lemma regardless of how the reform margins resolve.

The milestone mapping is fixed. March 2027 is the milestone talk, not the draft: it requires the first-stage facts, the exposure measure, the model skeleton, and an anatomy pilot. The full draft follows in summer 2027, serving the autumn market. Slippage triggers pre-agreed deletions — chronologies, the signalling block, the full deal-universe anatomy, the pre-2022 backfill, and finally the EDGAR-only fallback — rather than schedule denial.

7 The Next Eight Weeks: Gates and Cut Order

The schedule below is deliberately gate-first. The model section of this outline is a provisional lens, already drafted to generate the exposure measure and the MDE arithmetic; it is frozen as-is until Gate 2. No *further* model investment and no price-dependent work is scheduled before the gates that decide whether those investments pay. March 2027 is a *milestone talk* (first-stage facts, the exposure measure, the model skeleton, an anatomy pilot), not a draft deadline; the full draft targets summer 2027.

Gate 0 (week 1): advisor ratification

A two-page decision memo (compressed from the verified decision record) and a supervision meeting. Every other gate is conditional on this one. The memo states plainly what is being abandoned (the liquidity-hump headline as the job-market paper), what is being frozen (the current 92-page draft, as a separate working paper after a bounded hygiene pass), and what is being built.

Gate 1 (weeks 1–2): access and hygiene, in parallel

1. **WRDS/CRSP go/no-go.** Account request submitted in week 1, calendar-booked. CRSP gates the exposure denominator (ADV), all return-based outcomes, and the anatomy leg. If access fails, the paper degrades along the published cut order below — it does not stall.
2. **Bloomberg terminal inventory** (`lit/bloomberg_checklist.md`): M&A deal fields, premia, export caps, academic terms of use. Decides the M&A-outcomes route (Bloomberg vs. hand-built EDGAR deal universe).
3. **Hygiene pass on the frozen draft** (bounded, ≤ 2 weeks, parallel work): the enumerated line-level corrections from the technical verification (cross-country threshold claims removed; the $\kappa \rightarrow 1$ narrative corrected; welfare units rescaled; the two misattributed citations fixed), then freeze.
4. **Gaussian-noise falsification** of the old model’s hump (one week, closes the question the pivot memo raised; result goes in the frozen draft’s appendix, not this paper).

Gate 2 (weeks 2–4): the pre-registered EDGAR pass (price-free)

Stake-at-filing distributions (untrimmed, split by corporate-action status), entry counts and composition by the coarse required-trading-days proxy, Item 6 derivative-language rates, 13G→13D switching, each with a minimum detectable effect computed *ex ante* from the SEC’s own tables.

Rule: filing-delay compression and post-deadline pre-filing accumulation are mechanical identities and cannot pass this gate. If at least one non-mechanical margin moves, proceed to the full build; if none moves, the model shrinks to the accounting lemma plus the corner condition, and the paper re-aims at the anatomy and the well-measured policy null.

Weeks 5–8: conditional build (critical path only)

Critical path: the exposure measure on CRSP ADV (or the EDGAR proxy if CRSP is delayed) and the trigger-date parser fix with hand-verification against Item 5(c) tables — the data map’s own component estimates for these are 1–2 and 2–3 weeks. The Item 4 purpose codebook on the 200-filing validated subset (3–5 weeks by the same map) and the anatomy pilot on the hand-collected acquired-target subsample *start* in this block only if the parser clears early, and otherwise roll past week 8: the eight-week promise covers the first-stage facts and a locked exposure design, not the codebook. Refinement of the model section beyond the frozen Section 4 specification proceeds *only if* Gate 2 passed on a behavioral margin.

Published cut order (if the schedule slips)

(1) merger-background chronologies stay out (no tooling exists in this repository — revisit only if tooling exists elsewhere); (2) the signaling block stays out (accounting lemma + corner model ship without it); (3) the full deal-universe anatomy shrinks to the hand-collected acquired-target subsample; (4) the 2010–2021 backfill is dropped (the SEC’s published tables anchor the pre-period); (5) if CRSP is still unresolved at week 4, the EDGAR-only configuration (stake, timing, composition, derivatives) is the paper.

Scoop defense

The novelty scan of 2026-08-02 found no realized-outcome study of the reform; the field is attentive and the data are public. First-stage facts and the design go up as an SSRN working paper as soon as the exposure design is locked (target: month 4–5), and the novelty scan and the Celentano–Levine publication status are re-checked at every revision.

References

- Albuquerque, Rui, Vyacheslav Fos, and Enrique Schroth (2022). “Value Creation in Shareholder Activism”. In: *Journal of Financial Economics* 145.2, pp. 153–178.
- Back, Kerry et al. (2018). “Activism, Strategic Trading, and Liquidity”. In: *Econometrica* 86.4, pp. 1431–1463.
- Betton, Sandra et al. (2014). “Merger Negotiations with Stock Market Feedback”. In: *Journal of Finance* 69.4, pp. 1705–1745.
- Bonetti, Pietro, Miguel Duro, and Gaizka Ormazabal (2020). “Disclosure Regulation and Corporate Acquisitions”. In: *Journal of Accounting Research* 58.1, pp. 55–103.

- Boyson, Nicole M., Nickolay Gantchev, and Anil Shivdasani (2017). “Activism Mergers”. In: *Journal of Financial Economics* 126.1, pp. 54–73.
- Brav, Alon et al. (2008). “Hedge Fund Activism, Corporate Governance, and Firm Performance”. In: *Journal of Finance* 63.4, pp. 1729–1775.
- Burkart, Mike and Samuel Lee (2022). “Activism and Takeovers”. In: *Review of Financial Studies* 35.4, pp. 1868–1896.
- Celentano, Francesco and Oliver Levine (2025). “Shareholder Activism, Takeovers, and Managerial Discipline”. R&R, Review of Financial Studies. Swiss Finance Institute Research Paper No. 25-81; SSRN 5506659.
- Cetemen, Doruk et al. (2026). “Leader-Follower Dynamics in Shareholder Activism”. In: *Journal of Finance*. Forthcoming.
- Collin-Dufresne, Pierre and Vyacheslav Fos (2015). “Do Prices Reveal the Presence of Informed Trading?” In: *Journal of Finance* 70.4, pp. 1555–1582.
- (2016). “Insider Trading, Stochastic Liquidity, and Equilibrium Prices”. In: *Econometrica* 84.4, pp. 1441–1475.
- Corum, Adrian Aycan and Doron Levit (2019). “Corporate Control Activism”. In: *Journal of Financial Economics* 133.1, pp. 1–17.
- Eckbo, B. Espen, Nadya Malenko, and Karin S. Thorburn (2025). *Corporate Takeovers*. ECGI Finance Working Paper No. 1030/2025. European Corporate Governance Institute.
- Edmans, Alex, Vivian W. Fang, and Emanuel Zur (2013). “The Effect of Liquidity on Governance”. In: *Review of Financial Studies* 26.6, pp. 1443–1482.
- Edmans, Alex and Clifford G. Holderness (2017). “Blockholders: A Survey of Theory and Evidence”. In: *The Handbook of the Economics of Corporate Governance*. Ed. by Benjamin E. Hermalin and Michael S. Weisbach. Vol. 1. North-Holland, pp. 541–636.
- Fishman, Michael J. (1988). “A Theory of Preemptive Takeover Bidding”. In: *RAND Journal of Economics* 19.1.
- Gantchev, Nickolay (2013). “The Costs of Shareholder Activism: Evidence from a Sequential Decision Model”. In: *Journal of Financial Economics* 107.3, pp. 610–631.
- Greenwood, Robin and Michael Schor (2009). “Investor Activism and Takeovers”. In: *Journal of Financial Economics* 92.3, pp. 362–375.
- Grossman, Sanford J. and Oliver D. Hart (1980a). “Disclosure Laws and Takeover Bids”. In: *Journal of Finance* 35.2, pp. 323–334.
- (1980b). “Takeover Bids, the Free-Rider Problem, and the Theory of the Corporation”. In: *Bell Journal of Economics* 11.1, pp. 42–64.
- Johnson, Travis L. and Nathan Swem (2021). “Reputation and Investor Activism: A Structural Approach”. In: *Journal of Financial Economics* 139.1, pp. 29–56.
- Kahn, Charles and Andrew Winton (1998). “Ownership Structure, Speculation, and Shareholder Intervention”. In: *Journal of Finance* 53.1, pp. 99–129.
- Kyle, Albert S. (1985). “Continuous Auctions and Insider Trading”. In: *Econometrica* 53.6, pp. 1315–1335.
- Maug, Ernst (1998). “Large Shareholders as Monitors: Is There a Trade-Off between Liquidity and Control?” In: *Journal of Finance* 53.1, pp. 65–98.
- Ordóñez-Calafi, Guillem and Dan Bernhardt (2022). “Blockholder Disclosure Thresholds and Hedge Fund Activism”. In: *Journal of Financial and Quantitative Analysis* 57.7, pp. 2834–2859.

- Polk, Ryan Christopher et al. (2024). “Shrinking the 13D Disclosure Window Will Benefit Non-Activist Investors”. In: *Journal of Financial Regulation and Compliance* 32.4, pp. 516–538.
- Schwert, G. William (1996). “Markup Pricing in Mergers and Acquisitions”. In: *Journal of Financial Economics* 41.2, pp. 153–192.
- Securities, U.S. and Exchange Commission (Oct. 2023). *Modernization of Beneficial Ownership Reporting*. Release Nos. 33-11253; 34-98704. Adopted October 2023; effective February 5, 2024. U.S. Securities and Exchange Commission. URL: <https://www.sec.gov/files/rules/final/2023/33-11253.pdf>.
- Securities, U.S. and Division of Corporation Finance Exchange Commission (2025). *Exchange Act Sections 13(d) and 13(g) and Regulation 13D-G Beneficial Ownership Reporting Compliance and Disclosure Interpretations*. Question 103.12 (Feb. 11, 2025; page last updated July 9, 2026). URL: <https://www.sec.gov/divisions/corpfin/guidance/13d-g-interps.htm> (visited on 08/02/2026).